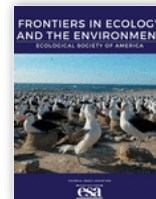


FRONTIERS IN ECOLOGY and the ENVIRONMENT

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Frontiers in Ecology and the Environment Author Guidelines

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- [General Information](#)
- [General Submission Criteria](#)
 - [Cover Letters](#)
 - [List of Authors](#)
 - [No Prior Publication](#)
 - [Titles and Abstracts](#)
 - [“In a nutshell” Section](#)
 - [Length of all Submissions](#)
 - [Where to Submit a Manuscript](#)
- [Categories of Papers Published in Frontiers](#)
 - [Research Communications](#)
 - [Concise Reviews](#)
 - [Concepts and Questions](#)
 - [Reader's Forum](#)
- [Frontiers' Peer-Review Process](#)
- [ESA Code of Ethics and Conflicts of Interest](#)
- [Legally Protected Species and Animal Welfare Certifications](#)
- [Authorship vs. Acknowledgements](#)
- [Open Research Policy](#)
- [Artificial Intelligence \(AI\) Policy](#)
- [Field Stations, Marine Laboratories \(FSMLs\), and Related Research Institutions](#)
- [Writing Style](#)
- [Formatting Your Manuscript for Online Submission](#)
- [Figures, Tables, and Panels](#)
 - [Figures](#)



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For Authors

- [Submit an article](#)
- [Calls for Papers](#)
- [Author Guidelines](#)
- [ESA Open Research Policy](#)
- [Publicity & Embargo Policy](#)
- [Permissions](#)

- [Figure Layout](#)
 - [Figure Captions and Credits](#)
 - [Tables](#)
 - [Panels](#)
 - [Reference Section](#)
 - [Supporting Information](#)
 - [Permissions](#)
 - [Licensing Overview](#)
 - [Data Repository Fees](#)
 - [Embargo Policy](#)
 - [Publication Fees – Option 1: Page Charges](#)
 - [Publication Fees – Option 2: Open Access Publication](#)
 - [Editorial Queries](#)
-

General Information

Frontiers in Ecology and the Environment publishes boundary-spanning papers that illustrate the importance of ecology and environmental science to scientific inquiry, biodiversity and ecosystem conservation, natural resource management, public policy, and other areas of ecological and environmental problem solving. *Frontiers'* hallmark is its readability and appeal to a wide audience in academia, government, and nongovernmental organizations (NGOs). We seek papers that (a) are written in clear, accessible language to maximize interest to professionals outside of an author's area of expertise; (b) integrate cutting-edge science with broad practical application, because such papers are thus well-positioned to impact professional practice on a regional or global scale; and (c) generate integrative solutions to challenges of our time, including but not limited to biodiversity conservation and climate change.

***Frontiers* enjoys a broad, global readership that includes:**

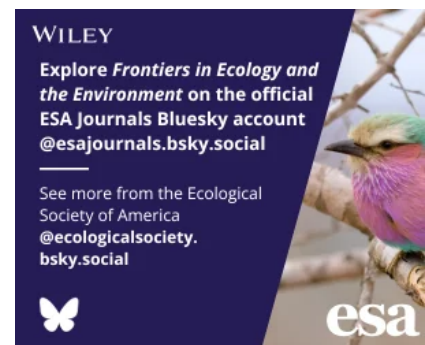
- Professional ecologists and environmental scientists in academia, government agencies, and NGOs
- Scientists in all subdisciplines of ecology and environmental science
- Resource managers, policy makers, educators, graduate and undergraduate students in ecology and environmental science, and
- Many other interested groups and individuals

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Frontiers has four criteria by which we evaluate the content of a submission:

1. Does the manuscript move applied science forward within its field of focus?
2. Will the paper's content be of interest to a diverse readership composed of scientists, practitioners, and educators in many fields?
3. Is the manuscript clearly and crisply written such that it will grab and hold the attention of readers regardless of their background, training, location, or field of practice?
4. Does the paper present an integrative analysis that clearly demonstrates its application to conservation, management, and/or policy?

The *Frontiers* editorial team embraces the Ecological Society of America's (ESA's) goals for diversity, equity, inclusion, and justice (DEIJ). At *Frontiers* we recognize the presence of institutionalized biases and the historical lack of diversity in scholarly publishing, including acute biases based on race, gender, geography, and age that have affected ecology and other life sciences. We therefore especially encourage the submission of manuscripts by historically underrepresented members of the scientific community and ask prospective authors to consider sharing relevant (anonymous) demographic information upon submission, which will help us broadly assess authorship in the context of achieving DEIJ goals.

When preparing and submitting manuscripts to *Frontiers* for potential consideration, prospective authors should simultaneously consider all the aforementioned information, as well as the following guidance below.

General Submission Criteria

Frontiers features four types of peer-reviewed papers:

- Research Communications
- Concise Reviews
- Concepts and Questions
- Reader's Forum

Below, you will find general guidance for all paper types followed by descriptions and criteria for submissions of each



Quantitative Analyst

Northwest Fisheries Science
Employer: A.I.S., Inc.

Assistant Professor of

Colorado Springs, Colorado
Employer: Colorado College

Endowed Professorship Position Management Mississippi State NWTF

Mississippi State University, MS
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individual paper type. Attention to *Frontiers'* author guidelines is the first criterion by which we assess the fitness of a manuscript for peer review, and manuscripts that are substantially out of compliance with *Frontiers'* author guidelines may be rejected without even a cursory review.

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Cover Letters

With each manuscript, authors should submit a cover letter that includes the following information (please do not simply cut and paste the abstract into your cover letter):

- Brief introduction to the work explaining its fit within *Frontiers'* aims and scope.
- Statement pertaining to authors' conflicts of interest (real or perceived), if any (see "Conflicts of Interest" section below)
- Declaration that the work being submitted is original and has not been previously published in any form, and has not simultaneously been submitted to nor is currently under consideration with any other journal or publishing outlet (see "No Prior Publication" section below)
- If an earlier version of the manuscript was submitted to *Frontiers*, but was rejected and invited for resubmission, provide the previous manuscript ID number and also a separate document detailing how you addressed reviewers' comments on the previous submission
- Very brief explanation of any other additional information or circumstances regarding the submitted manuscript that you believe will help with our initial review

List of Authors

Names of all authors and co-authors must be present in the online submission form *and* on the first page of the manuscript, beneath the article title. Also on the first page, state each author's affiliation(s), usually the institutional affiliation of the author during the period when all or most of the data were collected. Present address(es), if different, should be noted.

Generally, there can be only one corresponding author for a

manuscript (although exceptions are possible, granted on a case-by-case basis). However, the corresponding author during peer review is not required to be the corresponding author on the accepted manuscript (i.e., we allow changes to the listed corresponding author between acceptance and publication). Statements of author contributions may be provided within the paper's Acknowledgements section if you feel that such contributions should be specifically noted.

As *Frontiers'* content is designed to be succinct, we place limits on the number of authors on a paper due to space considerations. For all submissions except Reader's Forum articles, please limit the author list to 30 or fewer authors. We do grant exceptions in special circumstances but require that such circumstances be made clear in the cover letter accompanying the submission. Manuscripts submitted in the Reader's Forum category are limited to 15 authors, again with possible exceptions that must be sought in your cover letter. When we do grant exceptions for manuscripts with large numbers of authors, we reserve the right to place all but the first few authors and their affiliations in a supporting panel (see "Supporting Information" section below), in order to reduce the space occupied on the article's first page.

No Prior Publication

ESA journals require that all submissions be original contributions, with full disclosure of any possible redundant publication made in the accompanying cover letter (as mentioned above). Under certain circumstances, use of the same data in two or more publications is appropriate and beneficial. This may be particularly true when new information allows reinterpretation of previously published data. In many cases, however, duplicate publication is wasteful of journal space and user resources. Although it is the Editor's responsibility to decide whether specific duplications are useful, these decisions are generally based on information supplied by the authors. ESA journals have adopted a policy to facilitate this process. At the time of submission, authors must provide information describing the extent to which data or text in the manuscript have been used in other papers that are published, in press, submitted, or soon to be submitted (at *Frontiers* or elsewhere). In cases of overlap with other publications or submissions, authors should include copies of

said papers along with the current submission.

Sometimes it is difficult to assess whether a work has truly been published previously. Please err on the side of caution in considering overlap. Reference should be made to any closely related previous publication, especially if a table or figure is reproduced. If any data in a manuscript have been included in other published or unpublished manuscripts, the legend of each table or illustration reporting such data must cite those manuscripts. When in doubt, an author should supply copies of the previous publication; these copies will be shared with the associate editor and reviewers, who would then be asked to consider this matter. Authors are responsible for obtaining permission to reproduce previously published material (see “Permissions” section below).

A posting of an unpublished manuscript or thesis on an author’s personal or home institution’s website or ftp site generally will not be considered previous publication. Authors interested in posting a copy of their manuscript to a preprint server must obey the following guidelines: The version of the manuscript posted on the preprint server must be the original version submitted to the journal and cannot be any version improved by peer review. If the manuscript is accepted, the author must provide a link to the published paper with the preprint posting. Authors should disclose whether such a posting has been made at the time of submission. If a manuscript is available as part of a digital publication such as a journal, technical series, or some other entity to which a library can subscribe (especially if that publication has an ISSN or ISBN), we will consider that the manuscript has been published and is thus not eligible for consideration by *Frontiers*. Finally, a necessary test for prior publication is whether the author can legally transfer copyright to ESA.

Titles and Abstracts

Each article should begin with a title of no more than 95 characters (including spaces) and an abstract of about 150 words maximum. Titles and abstracts are among a paper’s most important features for engaging readers’ initial interest. Titles should be catchy, punchy, and avoid acronyms and jargon (scientific, technical, or discipline-specific terminology). Abstracts should do more than just repeat in brief what the article says; they must draw readers in by explaining what the

article is about and why it is important. This and the early introductory paragraphs of the main text should try to convey a sense of the enthusiasm that the writer feels for their subject.

“In a nutshell” Section

Two of *Frontiers’* article types, *Concise Reviews* and *Concepts and Questions*, contain both a traditional abstract and, immediately following, a section entitled “In a nutshell.” The abstract and “In a nutshell” section have different purposes, so the language and style used to write them should be different. “In a nutshell” is a brief bullet-point list provided for the benefit of non-scientific readers – particularly resource managers, students, the media, and informed members of the public. It should consist of 3–4 bullet points succinctly describing the background information and the main take-home messages of the paper in the clearest language possible. There should be no jargon (scientific, technical, or discipline-specific terminology), and no repetition from the abstract or main text. The total length (combined word count of all bullet points) should not exceed 100 words.

Length of all Submissions

Frontiers’ content is designed to be succinct, which places strict limits on manuscript length. *Frontiers’* editors therefore reserve the right to reject overlong manuscripts without review and to modify all submitted manuscripts – or require authors to do so upon revision or prior to publication – in order to conform to journal style and available space. If a submitted manuscript exceeds word and/or reference limits, or if space restrictions otherwise prohibit inclusion of submitted figures and/or tables, the editors also reserve the right to convert figures or tables to Supporting Information. Whenever we determine that a modification is necessary, we will communicate with authors prior to making any changes.

Where to Submit a Manuscript

All manuscripts (including letters) must be submitted through

Frontiers' online manuscript submission system, **ScholarOne**.

Prior to submission, and if you have not already done so, please check to see whether you are already in the ScholarOne database by entering your email address in the E-mail Address field under Password Help and click "Go". A record for you may exist in the database even if you were never previously an author of a manuscript submitted to any of the ESA journals, including *Frontiers*. Do not create a new account if you are already in the database. If you have verified that there is no account for you, you can create an account at the submission site by clicking on the "Register Here" button. From that point, the system should guide you through the submission process.

If you need help with your ScholarOne account or if you have any submission-related questions, please contact the assistant editor at esajournals@esa.org.

Please make electronic backup copies of everything that you have uploaded to ScholarOne.

Categories of Papers Published in *Frontiers*

Following is guidance specific to each peer-reviewed paper category:

Research Communications

Research Communications are relatively short, novel, high-impact papers that showcase primary research and clearly demonstrate the importance and application of ecological and related environmental science to conservation, management, or policy making. Authors interested in submitting a Research Communication should prepare a manuscript that meets these criteria:

1. The main text of Research Communications should always be divided into standard sections: Introduction, Methods, Results, and Conclusions (or a close variation thereof).
2. We seek submissions that are written in a style that is

crisp, concise, and accessible not only to ecologists and environmental scientists but also to those in management, policy, and education. Therefore, you should strive to avoid jargon (scientific, technical, or discipline-specific terminology) that might be unfamiliar to a broad, multidisciplinary readership. Similarly, please avoid using too many acronyms and be sure to spell out the meaning of any you use upon first appearance.

3. Content: initial submissions should
 - a) Not exceed **2500 words** in the main text (i.e., excluding abstract, title, author names/affiliations, references, figure legends BUT including any panel text, if applicable)
 - b) Contain a maximum of **25 references**
 - c) Include no more than **three (3) of any combination of the following components: figures, tables, and/or panels**
 - d) Include an **abstract with a maximum of 150 words**
 - e) Provide a **statement of compliance** with ESA's Open Research Policy
4. In addition, authors may submit a **maximum of five (5) components** of Supporting Information (including supporting figures, supporting tables, and supporting panels). See "Supporting Information" section below for additional details.

Concise Reviews (formerly Reviews)

Concise Reviews are synthesis papers that may focus on a wide range of ecological and environmental science topics but are expected to feature a strongly interdisciplinary and integrative analysis designed to advance the paper's field of focus. As with all paper types in *Frontiers*, Concise Reviews should address as a central concern practical application to conservation, management, or policy making. Topics should have broad, interdisciplinary appeal. Authors interested in submitting a Concise Review should prepare a manuscript that meets these criteria:

1. We seek submissions that are written in a style that is crisp, concise, and accessible not only to ecologists and environmental scientists but also to those in management, policy, and education. Therefore, you should strive to avoid jargon (scientific, technical, or

discipline-specific terminology) that might be unfamiliar to a broad, multidisciplinary readership. Similarly, please avoid using too many acronyms and be sure to spell out the meaning of any you use upon first appearance.

2. Content: initial submissions should
 - a) Not exceed **3000 words** in the main text (i.e., excluding abstract, title, author names/affiliations, references, figure legends BUT including any panel text, if applicable)
 - b) Contain a maximum of 50 references
 - c) Include no more than **six (6) of any combination of the following components: figures, tables, and/or panels**
 - d) Include an **abstract with a maximum of 150 words**
 - e) Include an **“In a nutshell” section with a maximum of 100 words**
 - f) Provide a **statement of compliance** with ESA’s Open Research Policy
3. In addition, authors may submit **a maximum of five (5) components** of Supporting Information (including supporting figures, supporting tables, and supporting panels). See “Supporting Information” section below for additional details.

Concepts and Questions

Concepts and Questions papers bear some similarity to Concise Reviews (as described above), but showcase ideas not yet widely accepted or demonstrated by the scientific community – i.e., they may be more conceptual and less well developed in the literature. *Frontiers* expects submissions in this category to demonstrate the same approach to genre-expanding integrative analysis and policy/management relevance as do Concise Reviews. Authors interested in submitting papers in the Concepts and Questions category should prepare a manuscript that meets these criteria:

1. We seek submissions that are written in a style that is crisp, concise, and accessible not only to ecologists and environmental scientists but also to those in management, policy, and education. Therefore, you should strive to avoid jargon (scientific, technical, or discipline-specific terminology) that might be unfamiliar to a broad, multidisciplinary readership. Similarly, please

avoid using too many acronyms and be sure to spell out the meaning of any you use upon first appearance.

2. Content: initial submissions should
 - a) Not exceed **2800 words** in the main text (i.e., excluding abstract, title, author names/affiliations, references, figure legends BUT including any panel text, if applicable)
 - b) Contain a maximum of **40 references**
 - c) Include no more than **four or five (4–5) of any combination of the following components: figures, tables, and/or panels**
 - d) Include an **abstract with a maximum of 150 words**
 - e) Include an **“In a nutshell” section with a maximum of 100 words**
 - f) Provide a **statement of compliance** with ESA’s Open Research Policy
3. In addition, authors may submit **a maximum of five (5) components** of Supporting Information (including supporting figures, supporting tables, and supporting panels). See “Supporting Information” section below for additional details.

Reader's Forum (formerly Letters)

Reader's Forum articles are very short, undergo peer review, and (at this time) only take the form of responses to articles that previously appeared in the journal. These submissions should provide additional or alternative interpretations of or perspectives on the original work. We do not publish Reader's Forum articles that constitute personal attacks on or airing of grievances between competing researchers, and *Frontiers'* editors reserve the right to assess the tone or appropriateness of any language or content in any submitted article.

Authors interested in submitting a Reader's Forum article should prepare a manuscript that meets these criteria:

1. As with other sections of the journal, it is crucial that the language contained within Reader's Forum articles be as clear and accessible as possible. To that end, authors should avoid jargon or specialized terminology wherever possible and focus on writing for the benefit of the non-specialist reader. Similarly, please limit use of acronyms to as few as possible (or none!) and be sure to spell out the meaning of any you use upon first appearance.

2. Content: initial Reader's Forum submissions should:
 - a) Include a brief title of no more than **65 characters with spaces**
 - b) Limit the main text (excluding title, author names/affiliations, references, and [if applicable] a figure legend) to no more than **800 words**
 - c) Contain a maximum of **12 references**
 - d) Include not more than **one (1) small figure or table**
3. In addition, authors may submit **a maximum of two (2) components** of Supporting Information (including supporting figures, supporting tables, and supporting panels). See “Supporting Information” section below for additional details.

For accepted Reader's Forum articles, the journal will invite a reply from the author(s) of the original work. If that author submits a reply, *Frontiers* will run both articles together in the same issue, but that will end the discussion – i.e., no further correspondence on the article in question will be accepted or published.

***Frontiers'* Peer-Review Process**

At *Frontiers*, all submitted manuscripts follow this process:

1. The author submits a manuscript to *Frontiers'* ScholarOne online submission system for consideration. Upon initial processing, the manuscript will be assigned a unique ID and the author will receive a notification of receipt via email.
2. The Editor in Chief (EiC) evaluates whether the manuscript will proceed to peer review. For manuscripts not advanced into review, the EiC informs the authors of this via email.
3. Manuscripts advanced into review are assigned to an associate editor (AE) for handling.
4. The AE reviews the manuscript and either recommends that it be rejected (in which case the EiC is notified; see step 9) or recommends several potential reviewers (see step 5).
5. The assistant editor invites the potential reviewers via email through ScholarOne.
6. Potential reviewers accept or decline (or fail to respond to) the invitation to review. Those that fail to respond in

due course automatically have their invitations withdrawn and then new potential reviewers are identified by the AE.

7. Once secured, the reviewers (usually two, sometimes three) typically submit their comments within 14 days, unless otherwise arranged with *Frontiers*' staff. This timeline may vary based in part on reviewers' availability.
8. Once all reviews are received, they are read and synthesized by the AE, who adds comments of their own and submits a recommendation to the EiC. AE recommendations are one of the following: reject, major revision, minor revision, or accept.
9. On the basis of the reviewers' comments and AE's recommendation, the EiC makes a final decision (reject, major revision, minor revision, or accept) and emails a decision letter to the author via ScholarOne, forwarding all of the review comments and any additional editorial guidance. The AE and reviewers remain anonymous throughout *Frontiers*' single-blind peer-review process.
10. Where major or minor revisions have been requested, the authors are asked to revise their manuscripts and resubmit them via ScholarOne, following the instructions provided by the EiC. The original AE is normally reassigned (pending their availability) to the revised manuscript, and may select the original reviewers, new reviewers, or a combination of both. The process begins again from step 5 and proceeds (for as many iterations as necessary) until the manuscript is either rejected or accepted.

ESA Code of Ethics and Conflicts of Interest

All authors submitting manuscripts to *Frontiers* agree to abide by the ESA's **Code of Ethics**, which addresses issues including (but not limited to) authorship, plagiarism, fraud, unauthorized use of data, copyrights, errors, confidentiality, intellectual property, attribution, willful delay of publication, and conflicts of interest, as well as other matters that are not specific to the publication process. All *Frontiers* editors also adhere to the ESA Code of Ethics.

If ESA has reason to doubt the ethical practices of an author of a manuscript, either because of concern raised by an editor, or

because of information obtained from some other source, the EiC will process the manuscript in accordance with normal practice, but will simultaneously refer the matter to the ESA Professional Ethics Committee for review. The Committee will conduct whatever investigation it feels appropriate, taking care not to inadvertently damage the reputation of any of the parties concerned. The EiC will receive the advice of the Committee and decide a course of action in consultation with the Executive Director of the Society.

We require all authors to declare any conflicts of interest in the relevant space on the online manuscript submission form and in the submission's cover letter. We ask that all authors disclose financial and personal relationships with other persons or organizations that could inappropriately influence (bias) their work. Examples of financial conflicts include employment, consultancies, stock ownership, honoraria, paid expert testimony, patent applications, and travel grants occurring within three years of beginning the work submitted. If there are no conflicts of interest, authors should state that there are none. Acknowledgements, including relevant sources of funding, should be declared in a brief statement at the end of the manuscript text. (For more information, see the Professional Practice section of the ESA's **Code of Ethics**.)

Legally Protected Species and Animal Welfare Certifications

Authors whose reported research involves species protected under relevant federal law (e.g., US Endangered Species Act, US Marine Mammal Protection Act) must certify that their research was conducted pursuant to applicable permits in compliance with the relevant laws.

Authors using experimental vertebrate animals must certify that their care was in accordance with institutional guidelines.

Authorship vs. Acknowledgements

Individuals listed as authors should have played a significant role in designing or conducting the research, writing the

manuscript, or providing extensive guidance on the execution of the project. Those whose role was limited to providing materials, financial support, or review should be recognized in the Acknowledgements section.

It is inappropriate to name multi-author collaborative and consortia groups as authors. If the entire group is entitled to authorship, every individual should be listed specifically.

Open Research Policy

Authors are required to follow the guidelines defined in the **ESA Open Research Policy**. Please review the policy before preparing a manuscript for submission. Your Open Research statement should be prepared before starting the submission process and included on the title page of the manuscript. Please be prepared to copy your finalized Open Research statement into the corresponding section of the online submission form.

To further ESA's commitment to Open Science and **FAIR** (Findable, Accessible, Interoperable, and Reusable) principles, two Data Editors are available to guide authors and ensure compliance with ESA's Open Research Policy. If you have any questions regarding how to meet data and code archiving policies while preparing your submission, please send an email to esajournals@esa.org and a Data Editor will respond.

Artificial Intelligence (AI) Policy

ESA Publications follow the **Committee on Publication Ethics (COPE)** position that artificial intelligence (AI) tools cannot be recognized as authors, as they cannot take responsibility for the submitted manuscript. If authors use AI tools such as ChatGPT or other large language models in the writing of a manuscript, in the production of images, or in the collection and analysis of data, they must transparently disclose how and which tool was used in the section of the manuscript to which use of AI pertains, with an additional disclosure statement in the paper's Acknowledgments. In addition, this must be disclosed in the manuscript submission form. Tools that are

used to improve spelling, grammar, and general editing are excluded from the scope of these guidelines. Authors are responsible for all content generated by AI tools.

Field Stations, Marine Laboratories (FSMLs), and Related Research Institutions

In recognition of the critical role that field stations, marine labs (FSMLs) and related research institutions and facilities serve in ecology, we ask that authors who have conducted research at or with the support of FSMLs provide the names and locations of FSMLs in both the Methods and Acknowledgments sections of the manuscript. By asking authors to standardize how they report the support provided by FSMLs, readers will more easily and accurately understand where data were collected.

Writing Style

As stated above, *Frontiers'* readership includes a broadly multidisciplinary, international community of scientists and environmental professionals from a wide range of backgrounds. Therefore, *Frontiers'* writing style is less “traditionally academic” than many other journals. It is crucial that the language contained within your manuscript be as clear and accessible as possible, especially with respect to complex or technical content. Please minimize the use of jargon (scientific, technical, or discipline-specific terminology) wherever possible in the text (and explain it clearly when you do use it). The following guidelines apply to both the abstract and the main body of the text:

- Briefly and clearly explain technical concepts and terms when they are first mentioned
- Where possible, choose commonly used words and more accessible language rather than complex technical terms
- Use the active instead of the passive voice
- Use shorter sentences rather than longer ones
- Write concisely, avoiding fillers (e.g., “The fact that” and “In order to”)
- Break the text up into paragraphs of no more than a few

hundred words, and add short, imaginative subsection headings whenever the subject changes, or simply to avoid overlong blocks of uninterrupted text

- Give the scientific name (genus and species, in italics and in parentheses) of each species the first time it is mentioned, in addition to its common name, if any. The common name can then be used throughout the text. Genus names should start with a capital letter and may be abbreviated to a single letter after their first mention, if no confusion will result
- Use as few acronyms as possible, and spell out all acronyms at first mention, putting the acronym in parentheses and using it from then on
- Include specific details such as names, places, and numbers when appropriate and if permitted
- Ask yourself if your piece will be interesting and clearly understandable to readers in areas of professional practice different than your own, including to nonscientists in resource management and conservation fields. If in doubt, consider sharing your article with colleagues in other departments and seeking feedback
- *Frontiers* does not use footnotes; work the information into the text or include as a panel (or supporting panel)
- If you use automatic citation software such as Reference Manager, Endnote, or Zotero, remove all field shading before submitting your manuscript
- Always add page numbers and line numbers to the manuscript prior to submission

Formatting Your Manuscript for Online Submission

We prefer manuscripts to be prepared and submitted in Microsoft Word (.docx) format, if possible. Please do not submit manuscripts as PDFs, as it interferes with the process of initial review. (ScholarOne will automatically create a complete PDF from the separate pieces of your submitted manuscript.)

Tables and figures must *not* be embedded within the main text. Figures should be prepared as individual high-resolution files (preferably in jpeg or tiff format) and uploaded separately from the main text of the manuscript. Figure *captions* and any tables should be placed at the end of the paper, after the references section. It is not necessary to specify where tables and figures

should be placed in the main text, provided that each of these elements has at least one in-text callout (e.g., “Figure 1”, “Table 1”, etc.). In-text callouts for figures, tables, and panels must appear in sequential/numerical order (e.g., the first callout for Figure 1 must appear earlier in the text than the first callout for Figure 2).

See the following section for more information on preparing figures, tables, and panels.

Figures, Tables, and Panels

As is the norm in scholarly publishing, *Frontiers*’ papers are made much more appealing to our readers when they include a thoughtful selection of pertinent figures, tables, and/or panels. See each manuscript type’s specific guidance, above, for respective limits on the allowable number of figures, tables, and/or panels. Below is our guidance on the preparation of these materials for submission with your manuscript. Important: if you intend to include previously published figures or other materials with your submission, please refer to the “Permissions” section below for additional details.

Note that figures, tables, and/or panels cannot take up a full page of text on a page, and so must be sized appropriately.

Figures

High-quality color graphics and high-resolution, high-quality color photos are strongly encouraged for *Frontiers* papers. There are no charges for use of color in figures in *Frontiers*.

Each photo and figure should be submitted as a separate file (jpeg or tiff files preferred) in the file list. The file name should consist of the lead author’s last name and the figure number (and letter, where part of a multipart/multipanel figure [e.g., Smith Figure 2a]). If photographs are embedded within a table or figure, please supply high-resolution versions of those photos.

The following are notes on styles and fonts required for figures appearing in *Frontiers*.

- Always use Arial or Helvetica as the font for ALL in-figure text, keys, and axis labels, etc.
- Graphics should be in color wherever possible, on a white background. When the same type of information is presented in more than one figure, colors should be used consistently across all figures (e.g., if red represents temperature in Figure 1, then that same color red should be used in any subsequent figures that display temperature)
- If possible, please use color schemes that are distinguishable by those with color-blindness (e.g., avoid using reds and greens of the same intensity).
- Photos must be high resolution (300 pixels-per-inch [ppi] at a width of 3.5 inches [8.9 cm]; original file size at least 1.5–2 MB or more)
- For multipart figures, label each part using parentheses and lowercase lettering, preferably in the top-left corner (but always in the same position in each panel) – e.g.: (a), (b), and (c) – these should be in black or white font, to aid visibility; do not include opaque circles or squares under these panel designators
- For all in-figure text, keys, and axis labels, capitalize the first letter of the first word only – the rest should be lowercase, unless it is a place name or proper name, which would normally be capitalized
- Use American English spelling (as opposed to British English)
- Use SI units throughout; selected exceptions: hectares (ha), degrees Celsius (°C), metric tons (t), liters (L), seconds (s), minutes (min), hours (hr), years (yr)
- Except for accepted unit abbreviations such as above, spell out all words if adequate space is available (e.g., “Agriculture”, not “Agric”)
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- For values between 10 and 9999, do not separate numbers with commas or spaces
- For values greater than 9999, separate numbers in groups of three with a comma: 10,000, 100,000
- Follow journal style conventions for units in axis labels (e.g., replace “square kilometers per year” with “ $\text{km}^2 \text{ yr}^{-1}$ ”)
- Do not use periods in acronyms or abbreviations (no periods or commas in eg or ie; important exception: *et al.*)
- Individual panels within figures should not be set off by

boxes or other edging

- Do not forget to add axis labels and units to graphs. For maps, add scale bars and compass roses. For aerial, macro, or micro photographic images, add scale bars as appropriate
- Do not use grid lines in graphs
- Use en dashes (–), not hyphens (-), to indicate negative numbers, including those in sub/superscripts. Use the en dash also to separate ranges (e.g., 15–25 days; March–May)

Figure Layout

In laying out figures, keep things clear and simple and try to maximize the space given to the data. Avoid both clutter and wasted empty (white) space.

- Figures published within *Frontiers'* Research Communications, Concepts and Questions, or Concise Review paper categories are most often 3.5 or 4.5 inches (8.9 or 11.4 cm) wide, and are occasionally 6 inches wide; please create your original images as close to these dimensions as possible to avoid distortion caused by shrinking or enlarging, and to preserve the best resolution
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- Titles or labels not absolutely necessary for understanding the figure should be removed and/or explained in the figure legend
- Use of color for the graphic elements (lines, symbols, etc.) is welcomed, particularly where this helps readers to understand what is being illustrated
- Use solid (filled) symbols for plotting data if possible, unless data overlap or there are multiple symbols; make symbols large enough that they will be distinguishable when the figure is reduced
- Do not use three-dimensional graphics unless necessary
- Scales or axes should not extend beyond the range of the data plotted
- Standard line weight (thickness) is 0.5 points for boxes, graphs, etc., but this can be increased to up to 2 points for line graphs
- Keys to symbols should be kept as simple as possible

and be positioned so they do not needlessly enlarge the figure; details can be included in the captions

- Tick marks along the axes of graphs should point outwards from the axis (left of y-axis and below x-axis)

Figure Captions and Credits

- Each figure requires an explanatory caption; all figure captions should be listed in sequential order at the end of the paper, after the References section.
- Each caption should be under 100 words, and preferably under 50. Be clear and concise. Information in overlong captions should be integrated into the main text.
- We strongly discourage the use of images (e.g., photographs, icons, silhouettes, illustrative artwork) in figures beyond the data necessary to report your findings. The ESA editorial office reserves the right to remove any clipart/silhouette and/or require that the clipart/silhouette be replaced with a text label. Please review our Permissions section for details.

Tables

- Tables of reasonable size and sidebars (panels) containing extra information are also welcomed; in an effort to conserve space, very large tables may need to be converted into supporting tables (see “Supporting Information” section below)
- Do not embed tables within the main text of the paper; tables should be placed toward the end of the paper, after the References and figure captions
- Try to limit tables to 200 words and five columns; if you have more information than this, please consider, in order of preference, (1) trimming down the information, (2) dividing it into multiple tables (while adhering to the manuscript type-specific limits for content described above), or (3) contacting editorial staff for guidance on Supporting Information (see also below)

Panels

In *Frontiers*, a panel is a brief amount of showcased text (often in paragraph form but may also be displayed as a short list, such as a glossary) that catches the reader's eye and complements the main text by providing a more extensive description of a particular topic.

Panels should be less than 500 words, have a short but descriptive title, and include no more than one associated figure. Please be judicious with the amount of text you include in your panels and note that the word counts of panels *are included within the total word count limit of the main text of your manuscript*. (See each manuscript type's specific guidance, above, for respective word limits.)

In your submitted manuscript, panels should be placed toward the end of the paper, after the References, figure captions, and tables.

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For journal name abbreviations, *Frontiers* generally follows ISI Web of Science style ([see this guide](#)).

Following is style guidance for several different types of references:

Article in Journal

Belhabib D. 2021. Ocean science and advocacy work better when decolonized. *Nat Ecol Evol* **5**: 709–10.

Keeler BL, Chaplin-Kramer R, Guerry A, *et al.* 2017. Society is ready for a new kind of science – is academia? *BioScience* **67**: 591–92.

LaRue EA, Hardiman BS, Elliott JM, and Fei S. 2019. Structural diversity as a predictor of ecosystem function. *Environ Res Lett* **14**: 114011.

Lovell ST and Taylor JR. 2013. Supplying urban ecosystem services through multifunctional green infrastructure in the United States. *Landscape Ecol* **28**: 1447–63.

Book

Clewell AF and Aronson J. 2013. Ecological restoration: principles, values, and structure of an emerging profession (2nd edn). Washington, DC: Island Press.

Keeley JE, Bond WJ, Bradstock RA, *et al.* 2012. Fire in Mediterranean ecosystems: ecology, evolution and management. Cambridge, UK: Cambridge University Press.

Kimmerer RW. 2020. Braiding sweetgrass: Indigenous wisdom, scientific knowledge, and the teachings of plants. Minneapolis, MN: Milkweed Editions.

Chapter in Book

Hortle KG. 2009. Fisheries of the Mekong River Basin. In: Campbell IC (Ed). The Mekong. New York, NY: Academic Press.

May RM and Anderson RS. 1983. Parasite–host coevolution. In: Futuyama DJ and Slatkin M (Eds). Coevolution (3rd edn). Sunderland, MA: Sinauer.

Suding K, Spotswood E, Chapple D, *et al.* 2016. Ecological dynamics and ecological restoration. In: Palmer MA, Zedler JB, and Falk DA (Eds). Foundations of restoration ecology. Washington, DC: Island Press.

Conference Proceedings

Cherrett JM. 1989. Key concepts: the results of a survey of our members' opinions. Proceedings of the 31st Symposium of the British Ecological Society; 4–6 Apr 1989; Southampton, UK. London, UK: British Ecological Society.

Renwick WH, Sleezer RO, Buddemeier RW, and Smith SV. 2006. Small artificial ponds in the United States: impacts on

sedimentation and carbon budget. Proceedings of the Eighth Federal Interagency Sedimentation Conference; 2–6 Apr 2006; Reno, NV. Reston, VA: US Geological Survey.

Scientific and Technical Reports and their Parts

Grant GE, Lewis SL, Swanson F, *et al.* 2008. Effects of forest practices on peakflows and consequent channel response: a state-of-science report for western Oregon and Washington. Portland, OR: US Department of Agriculture. PNW-GTR-760.

Hock R, Rasul G, Adler C, *et al.* (Eds). 2019. High mountain areas. In: Pörtner H-O, Roberts DC, and Masson-Delmotte V, *et al.* (Eds). Special report on the ocean and cryosphere in a changing climate. Geneva, Switzerland: IPCC.

IPCC (Intergovernmental Panel on Climate Change). 2007. Climate change 2007: synthesis report. Contribution of Working Groups I, II, and III to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. Geneva, Switzerland: IPCC.

Dissertations and Theses

Feth JA. 1947. The geology of Northern Canelo Hills (PhD dissertation). Tucson, AZ: University of Arizona.

Osano PM. 2005. Estimating the opportunity costs of biodiversity conservation in Western Cape, South Africa (MS thesis). Cape Town, South Africa: University of Cape Town.

Websites

BirdLife International. 2018. Bird species distribution maps of the world (v2018.1). Cambridge, UK: BirdLife International. <http://datazone.birdlife.org/species/requestdis>. Viewed 23 Oct 2020.

Faust LJ, Theis M, Long S, and Shell S. 2011. PMCTrack: a website for monitoring breeding and transfer recommendations for zoo programs. Chicago, IL: Lincoln Park

Zoo. <https://www.pmctrack.org>. Viewed 12 May 2021.

PRISM Climate Group. 2020. PRISM climate data. Corvallis, OR: Oregon State University. <https://prism.oregonstate.edu>. Viewed 13 Aug 2020.

Newspaper Articles

Baker JK. 1999. Switch to dollar bodes ill for Ecuador. *Washington Post*. Sep 12.

Articles In Press*

Fulton RS. In press. Predator–prey relationships in an estuarine littoral copepod community. *Ecology*.

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Loss SR, Li BV, Horn LC, *et al.* 2023. Citizen science to address the global issue of bird–window collisions. *Front Ecol Environ* **21**: 418–27.

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Lamb JS, Cooper-Mullin C, Gilliland SG, *et al.* 2024. Evaluating conservation units using network analysis: a sea duck case study. *Front Ecol Environ* **22**: e2648.

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